AP PRECALCULUS

Unit 1 • Lesson 1.11 practice

Lesson 1.11 • Equivalent Representations of Polynomial and Rational Expressions

4 multiple-choice questions and 1 free-response question

Question 1

Which is the standard form of $(x+1)^3$?

(A) x^3+1

(B) x^3+3x^2+3x+1

(C) x^3+3x+1

(D) x^3+2x^2+2x+1

Question 2

What is the remainder when x^3-2x+5 is divided by $x-2$?

(A) 1

(B) 5

(C) 9

(D) 13

Question 3

Which is the completely factored form of x^3-9x?

(A) $x(x-3)(x+3)$

(B) $x(x-3)^2$

(C) $(x-3)(x+3)$

(D) $x(x^2+9)$

Question 4

What is the coefficient of x^3 in the expansion of $(x+2)^5$?

(A) 10

(B) 20

(C) 80

(D) 40

Question 5 (Free response)

Let $p(x)=2x^3+x^2-13x+6$.

(a) Show that $x=2$ is a zero of p.

(b) Write $p(x)$ as a product of linear factors and find all zeros of p.

(c) Write $\dfrac{p(x)}{x+1}$ in the form $q(x)+\dfrac{k}{x+1}$.